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11 January 1967
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4 December 1964

MILITARY SPECIFICATION

PACKING, PREFORMED, PETROLEUM HYDRAULIC FLUID RESISTANT, 275°F

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers petroleum hydraulic fluid resistant O-ring packings of the sizes shown on MS28775 and one grade only.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-A-250/4	Aluminum Alloy 2024, Plate and Sheet
QQ-B-626	Brass, Leaded and Nonleaded: Rods Shapes, Forgings, and Flat Products with Finished Edges (Bar, Flat Wire and Strip)
QQ-P-750	Bronze, Phosphor, Bar, Plate, Rod, Sheet, Strip, Flat Wire, and Structural and Special Shaped Section

Military

MIL-P-4861	Packing, Preformed, Rubber, Packing, Packaging of
MIL-P-5514	Packing, Installation and Gland Design, Hydraulic, General Requirements For
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile, and Ordnance
MIL-H-6083	Hydraulic Fluid, Petroleum Base Preservative
MIL-S-18729	Steel Plate, Sheet, and Strip, Alloy, 4130, Aircraft Quality

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STANDARDS

Federal

Fed. Test Method Std. No. 601	Rubber: Sampling and Testing
Fed. Test Method Std. No. 791	Lubricants, Liquid Fuels, and Related Products; Methods of Testing

Military

MIL-STD-10	Surface Roughness, Waviness, and Lay
MIL-STD-831	Test Reports, Preparation of
MS 28775	Packing O-ring, Hydraulic + 275°F

Air Force-Navy Aeronautical

AN6227	Packing: O-Ring Hydraulic
AN6230	Gasket: O-Ring Hydraulic

Publications

Air Force-Navy Aeronautical Bulletin No. 438	AGE Controls for Synthetic Rubber Parts

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society for Testing Materials Standard Method of Tests Designations

D91 - Method of Test for Precipitation Number of Lubricating Oils

D1329 - Evaluating Low Temperature Characteristics of Rubber and Rubber-Like Materials by a Temperature - Retraction Procedure (TR Test)

(Copies of ASTM publications may be obtained from the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

3. REQUIREMENTS

3.1 Qualification. The packings furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein and has been listed on or approved for listing on the applicable qualified products list.

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3.2 Material. The packings shall be manufactured from a synthetic rubber which is compatible with hydraulic fluid conforming to military specification MIL-H-5606. The synthetic rubber shall contain no substances which will adversely affect the properties of the fluid or packings.

3.3 Design and construction. The design and construction of the packings shall be in accordance with MS 28775, as applicable.

3.3.1 Splicing. Splicing of cured rubber shall not be used in the manufacture of the packings. Splicing of uncured rubber is allowable.

3.4 Physical Properties. Physical properties of the synthetic rubber shall conform to Table I.

TABLE I

Property	Characteristics
Original physical properties	
Specific gravity (points)	As determined.
Hardness at 75 ± 5 F (shore A durometer points).[2]	
Tensile strength (psi)	As determined.
Ultimate elongation (percent)	180 or higher.[1]
Tensile stress (modulus) at 100 percent elongation (psi)	As determined.
Permanent set (percent)	Not greater than 4 at 150 percent elongation.
Temperature retraction, 50 percent elongation and 10 percent return	As determined.
Tear resistance	As determined.
Permissible change in original physical properties after oil and air aging.	
Tensile strength (percent change.)	As determined.
Elongation (percent change)	As determined.
Hardness (change in shore durometer points).[2]	As determined.
Permanent set (list actual value) air aging only.	Not greater than 4 at 150 percent elongation.
Volumetric change after aging in hydraulic oil (percent).	0 to 15
Temperature retraction, 50 percent elongation and 10 percent return	-49°F or colder

[1] Tolerance would allow this to go down to 160 percent.

[2] Tests are to be made on hardness samples. Hardness shall not be determined from actual packings. All other tests are to be performed on packings as applicable.

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3.4.1 The maximum permissible variations in original physical properties during actual production from those values established during qualification tests shall be within the limits specified in Table II, but most stay within the limits specified in Table I.

TABLE II

Property	Characteristics
Maximum permissible production variations in physical properties from contractor's qualification values.[1]	
Specific gravity (points)	0.02[2]
Hardness at 75 ± 5 F (shore A durometer points).	±3[3]
Tensile strength (percent)	±15
Elongation (percent)	±20
Tensile stress (modulus), 100 percent pounds (percent)	±25
Permanent set (percent)	±50 up to 1.5 percent or ± 20 above.
Temperature retraction, 50 percent elongation and 10 percent return	± 3°F
Corrosion and adhesion	Slight on steel.

1 See Service Qualification Approval Report for limiting production values for a given compound.

2 If basic specific gravity is 1.30, then limits can be 1.28 to 1.32.

3 Shore hardness on production samples shall be not less than 65 for material with this tolerance applied.

3.5 Finish. Mold flash shall be removed from the packing in such a manner that they conform to the requirements specified herein and on applicable standards. All surfaces of the packings shall be reasonably free from defects, blemishes or irregularities. Reasonable shall be defined as allowing minor pits, raised spots, or tool marks which shall not be greater than 0.003 inch in height or depth from the packing surface.

3.6 Performance.

3.6.1 Leakage and breakout. Packings shall not exceed specified values when tested as described in Section 4. There shall be no evidence of wear, extrusion, deterioration, or excessive permanent set.

3.6.2 Cycling. Packings shall withstand the operation specified in Section 4. Leakage shall not exceed values specified in Section 4. After test, the packings shall be serviceable with no evidence of cracking, flaking, excessive wear, or other malfunctioning. Normal wear shall not be considered as malfunctioning.

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3.6.3 Fatigue. All packings when tested as specified in 4.6.10 shall perform without fatigue failure.

3.6.4 Crush. Material shall be capable of withstanding high-compressive loads without cracking or otherwise disintegrating, when tested as specified in 4.6.11.

3.7 Identification of product. MS 28775 packings shall have no identification on the packings. The manufacturer and specification shall be identified on the package.

3.8 Age. Unless otherwise specified, the age limitations for the finish product shall conform to ANA Bulletin, No. 438.

3.9 Workmanship. Workmanship shall be in accordance with the highest grade practice in manufacturing this kind of product. Manufacturing practices shall be such that the physical properties of the finished product shall be uniform through the cross section of the packing.

3.9.1 Packings may be off-register due to mold misalignment, not to exceed .003 inch provided all cross-sectional diameters, including the protrusions (flash excluded) of both halves, where applicable, satisfy cross-sectional dimensions.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Tests. The inspection and testing of packings shall be classified as follows:

a. Qualification Tests - - - - - - - - - - - 4.3

b. Quality Conformance Tests - - - - - - - - 4.4

4.3 Qualification Tests

4.3.1 Test samples. Qualification test samples shall consist of the following.

43	MS28775-214	Packings
56	MS28775-218	Packings
55	MS28775-325	Packings
3	ASTM hardness test disks (½ inch thick by 1 inch diameter).	

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4.3.1.1 Hardness test disks shall have the same compounding and equivalent cure as the accompanying packings, and the packings shall represent the manufacturer's production product, particularly with respect to stock preparation, mold design, and molding technique.

4.3.2 Tests. The qualification tests shall consist of tests for the following properties as specified under Test Methods (Para 4).

a. Temperature retraction (new and oil aged) (4.6.4.1, 4.6.4.2).
b. Crush (4.6.11).
c. Low-temperature performance "A" (4.6.6.1).
d. Cycling (aged and unaged +120°F and +275°F (4.6.7.1, 4.6.9.1, 4.6.9.2).
e. Fatigue (4.6.10).
f. High-temperature performance (4.6.8.1).
g. Specific gravity (new only) 4.6.2).
h. Hardness (new, air, and oil aged) (4.6.2, 4.6.3.1, 4.6.3.2.1).
i. Tensile strength (new, air, and oil aged) (4.6.2, 4.6.3.1, 4.6.3.2.1).
j. Ultimate elongation (new, air, and oil aged) (4.6.2).
k. Permanent set (new and air aged) (4.6.2.3, 4.6.3.1).
l. Corrosion and adhesion (new only) (4.6.5).
m. Volume change (oil aged) (4.6.3.2.1).
n. Tensile stress (modulus)(new and oil aged)(4.6.2).
o. Tear resistance (new only). (4.6.2.5).

4.4 Quality conformance tests. Quality conformance tests shall consist of the following:

a.	Examination of Product	4.6.1
b.	Batch tests	4.4.1
c.	Finished Product tests	4.4.2
d.	Process control tests	4.4.3

4.4.1 Batch tests. Batch tests shall consist of the following:

a.	Hardness	4.6.2
b.	Specific gravity	4.6.2
c.	Tensile strength	4.6.2
d.	Ultimate elongation	4.6.2
e.	Modulus @ 100% elongation	4.6.2

A batch shall consist of the quantity of material compounded at any one time, and ready for molding and curing into the finished product. Acceptance of the batch shall in no case be construed as a guarantee of the acceptance of the finished product.

4.4.1.1 Test specimens. Test specimens shall be molded from each batch and finished with the same state of cure as the finished product. Test specimens shall consist of the following:

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a. One hardness disc (0.25 inch thick and 1.00 inch diamter).
b. Three MS28775-218 packings.

4.4.1.2 Rejection and retest. Failure of any of the batch test specimens to conform to the requirements of this specification shall be cause for rejection of the batch represented. A batch that has been rejected may be reworked to correct the deficiencies and re-submitted for acceptance.

4.4.2 Finished product tests. The following tests shall be performed on unaged specimens:

a.	Specific gravity	4.6.2
b.	Tensile strength	4.6.2
c.	Ultimate elongation	4.6.2
d.	Tensile stress at 100% elongation	4.6.2

4.4.2.1 Test specimens. Three MS28775-218 packings or three MS28775-325 packings shall be selected at random from each day's production or shall be molded or cured concurrently with other production parts using the same production techniques. In the event of more than one work shift per 24 hour day, the three samples required shall be comprised of packings selected on an equal basis from the production of each work shift. All test specimens shall be furnished by the manufacturer.

4.4.2.2 Rejection and retest. Failure of any of the test specimens to conform to the acceptance requirements of this specification shall require a retest of an additional set of test specimens. In the event of failure of any of the retested specimens, the day's production of the finished product, represented by the specimens, shall be rejected. No further packings shall be accepted until the reason for failure has been determined and corrected and verified by reconducting the complete series of finished product tests (4.4.2).

4.4.3 Process control tests. Tests shall be conducted, as follows, at six-month intervals, on unaged specimens noted in 4.4.3.1:

a.	Specific gravity	4.6.2
b.	Hardness	4.6.2
c.	Tensile strength	4.6.2
d.	Ultimate elongation	4.6.2
e.	Tensile stress (modulus)	4.6.2
f.	Permanent set	4.6.2.3
g.	Temperature retraction	4.6.4.1
h.	Corrosion and adhesion	4.6.5
i.	Volume change	4.6.3.2.1
j.	Crush	4.6.11

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4.4.3.1 Test specimens shall be selected at random from production lots or shall be molded and cured concurrently with other production parts, using the same production techniques Test specimens shall consist of the following:

1 hardness disc (1/4 inch thick by one inch diameter minimum)
22 MS28775-214 packings
15 MS28775-218 packings
18 MS28775-325 packings

4.4.3.2 Rejection and retest. Failure of any of the test specimens to conform to the process control requirements of this specification shall require a retest of two additional sets of test specimens. In the event of failure of any of the retested specimens, all production items shall be rejected. No further packings shall be accepted until the reason for failure has been determined, corrected, and verified by reconducting the complete series of process control tests (4.4.3).

4.5 Test conditions. The test fluid shall be an oil which has been listed or approved for listing on QPL-5606, with 26-28% swell of standard synthetic rubber. The oil and samples to be aged shall be put into a nonpressurized closed-type container. A new batch of fluid shall be used for each aging test. In all cases, the ratio of quantity of fluid to the weight of the packings being aged shall be 20 ml of fluid for every gram of packings. Packings shall be loosely separated in the fluids.

4.6 Test methods.

4.6.1 Examination of product.

4.6.1.1 Each packing shall be carefully examined to determine workmanship, finish, trimming, marking, and conformance to the applicable standards.

4.6.1.2 Packings shall be checked for cross-sectional dimensions, ID, and OD. All dimensions shall conform to the applicable standard. Optical devices shall be used for measuring. There shall be no distortion of the material.

4.6.2 Original physical property of material. The physical properties listed below shall be determined as specified in Fed. Test Method Std. No. 601, Table I of this specification, and as noted. During Qualification tests only, the manufacturer shall report all values for each sample tested as well as the average value for each size:

a. Hardness (Method 3021).
b. Tensile strength (Method 4111).
c. Ultimate elongation (Method 4121).
d. Tensile stress (modulus) (Method 4131).
e. Permanent set (see 4.6.2.3).
f. Specific gravity (Method 14011).
g. Tear resistance (see 4.6.2.5).

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4.6.2.1 Hardness of the compound shall be determined as specified in 4.6.2, except that test samples shall be compression molded ¼ inch thick, and shall be no smaller than 1 inch in diameter. Two 1/8 inch sheets shall not be used. Hardness checks shall not be made on actual packings.

4.6.2.2 Tensile strength, Ultimate elongation, Tensile stress (modulus) and Permanent set for the material shall be made on actual qualification samples of packings for Qualification tests, and those quantities and sizes of samples tested shall be as specified in table III.

4.6.2.3 Permanent set. Permanent set tests for the material shall be made on packings as specified in table III, by measuring the ID and then stretching over a tapered cone lubricated with Specification MIL-H-5606 fluid until the ID of the packing has been stretched 150 percent. (Example: If ID of specimen measures 0.990 inch, the packing shall be stretched to 2.475 inches in diameter). The packing shall be held in this stretched state for 10 minutes, removed from the cone, and the new ID determined after 10 minutes rest. Permanent set shall be calculated from the as-measured ID before and after the test. Any breaking or cracking of the test samples shall be cause for rejection.

4.6.2.4 Specific gravity for the material shall be determined on the finished packings.

4.6.2.5 Tear resistance. Tear resistance shall be determined on packings for Qualification Tests only, except that at the option of the procuring activity this test may be performed on samples from a production lot when deemed necessary. The test shall be conducted on at least two samples (see Table III). In conducting this test, the packing shall be split on the centerline of the cross section in the manner indicated in Figure 1, and the tear test made in two planes, one with the mold seam and the second at right angles to the mold seam. A Scott test machine or equivalent may be used. Tear values recorded in the tear test shall be obtained with a 2-inch-per-minute rate of motion of lower jaw of the machine. (Table I). The tear resistance of the specimen shall be calculated as follows:

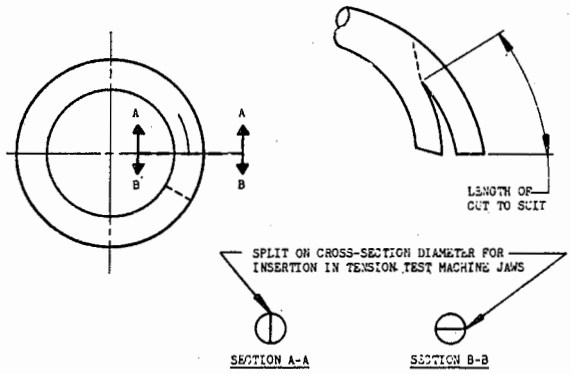


FIGURE 1.—*O-ring tear test details.*

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TABLE III. Test samples required for Qualification tests.

Aging Condition	Physical properties test for:	No. of samples for MS28775 Part No. -214	-218	-325
Unaged (table I)	Specific gravity.[1]			
	Tensile strength, Ultimate elongation, and Tensile stress (modulus).		6	6
	Permanent set.	2		
	Temperature retraction, 50 percent elongation and 10 percent return.		1	
	Tear resistance			2
	Corrosion and adhesion	5		
Aged (table I)	Tensile strength and elongation		6	6
	Permanent set	2		
Oil aged (table I)	Tensile strength, Elongation, and Tensile stress (modulus)		6	6
	Temperature retraction, 50 percent elongation and 10 percent return.		2	
	Volume change		9	9
Subtotal (physical properties)		9	30	29
	Performance and endurance tests:			
Aged and new	Performance and endurance	5	6	6
Unaged	Fatigue	2	2	2
Unaged and oil aged	Crush	9		
Subtotal (performance)		16	8	8
Additional samples required by Fed. Test Method Std. No. 601		18	18	18
Total (all tests)		43	56	55

[1] For the Specific gravity test, the data obtained from the packings designated for volume change prior to aging may be used.

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Tear resistance, pounds per inch of thickness = $\frac{F}{T}$

Where:
F= the maximum tearing force in pounds, and
T= the thickness of the specimen in inches.

4.6.3 Physical properties after aging.

4.6.3.1 Air aging. Air aging and testing of samples (see tables I and III) shall be conducted as specified in Fed. Test Method Std. No. 601, Method 7221. Hardness and permanent set shall be determined as specified in 4.6.2. The percentage difference between aged and unaged values for tensile strength and elongation (table I), and the actual values for change in hardness and permanent set (table I), shall be recorded.

4.6.3.2 Oil aging. The containers used for oil aging shall be of the conventional fruit-jar type. The cover for the container shall be either the two-piece (lid-and-ring) fruit-jar type, except that the lid shall be inverted in order that the lid's flat surface comes in contact with the sealing surface of the container, or a glass cover held firmly in place by a wire-lever apparatus. Unclean, uneven sealing surfaces of the cover and container and oil contact with the sealing surfaces shall be avoided. During oil aging the packings shall be suspended in the oil by a glass or nichrome wire rack in such a manner that the packings will not come in contact with any part of the container. Immediately after the aging period and before physical property determination, the packings shall be cooled in new fluid for a period of not less than 30 minutes at $77^{\circ} \pm 5^{\circ}F$. Oil aging shall be conducted in hydraulic oil conforming to Specification MIL-H-5606, as specified in 4.4, for a period of 72 hours at $275^{\circ} \pm 5^{\circ}F$ unless otherwise specified.

4.6.3.2.1 Tensile strength, elongation, volume change, and hardness. After aging as specified in 4.6.3.2, change in tensile strength and elongation shall be determined as specified in Fed. Test Method Std. No. 601, Method 6111; volume change shall be determined as specified in Fed. Test Method Std. No. 601, Method 6211; and change in hardness shall be determined in accordance with 4.6.2.1, except that, for all the determinations, qualification sample packings of sizes specified in table III shall be used, and the changes in physical properties shall be reported as described in table I. In addition, packings of sizes specified in table III shall be aged in hydraulic oil conforming to Specification MIL-H-5606 for a period of 7 days at $158^{\circ} \pm 2^{\circ}F$, and volume change only shall be determined and reported as specified above.

4.6.4 Temperature retraction.

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4.6.4.1 Unaged. This test shall be in accordance with ASTM D1329 except that MS28775-218 samples shall be used and 4 or 6 samples may be tested. The rate of warming shall be 1°F per minute. The temperature retraction 50/10 reading of these unaged packings shall be determined and recorded.

4.6.4.2 Aged. The test specified in 4.6.4.1 shall be repeated on applicable specimens. Two MS28775-218 packings shall be aged for 7 days in Specification MIL-H-5606 oil at a temperature of 158° ± 2°F. Compound having a temperature retraction 50-10 value of -49°F or colder, shall be subjected to a cold performance test. A suggested temperature retraction apparatus is shown in figure 2.

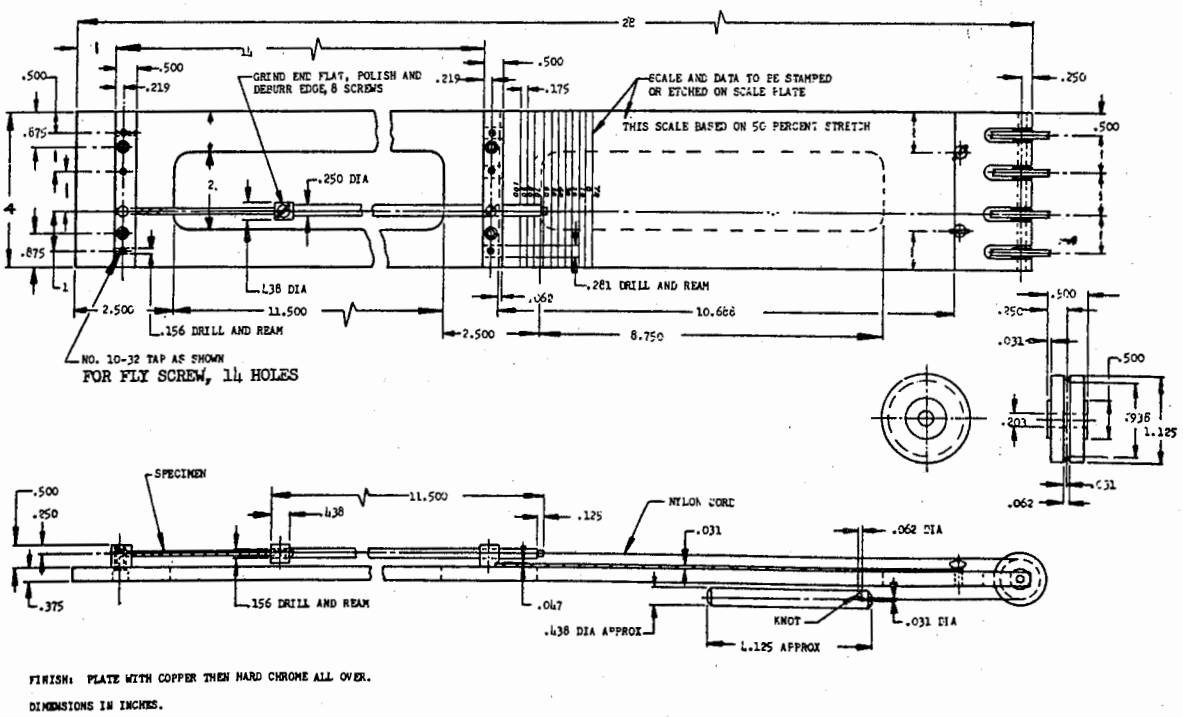


FIGURE 2.—*Temperature retraction apparatus.*

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4.6.4.3 Aging of the two MS28775 -218 packings for the temperature retraction test shall be accomplished by placing the packings in a small 4-ounce-screw-top jar and covering them with 56 ml of Specification MIL-H-5606 oil. A screw-type lid shall be placed snugly on the jar and the jar shall be put into an aging oven with the fluid temperature maintained at $158^{\circ} \pm 2^{\circ}F$ for 7 days.

4.6.4.4 For Quality control tests, only new unaged packings shall be used for the Temperature retraction test specified in 4.6.4.

4.6.5 Corrosion and adhesion. Packings of the type required for Qualification test (in quantities and sizes specified in table III) shall be prepared for corrosion testing by inserting sufficient quantities of the packings in a desiccator or similar humidity chamber maintained at 92 percent minimum relative humidity and room temperature for 72 hours minimum. Metallic plates of the metals listed below shall be polished to a surface roughness of 4 to 16 microinches RHR finish in accordance with Standard MIL-STD-10. The edges shall also be polished to reduce the formation of edge corrosion. Plates shall be washed with precipitation naphtha, as specified in Method 3101 of Fed. Test Method Std. No. 791, ASTM Method No. D91-52, or similar degreasing agent. The metals used shall be as follows:

a. Aluminum alloy; Specification QQ-A-250/4.
b. Brass; Specification QQ-B-626, composition 2.
c. Phosphor bronze; Specification QQ-P-750.
d. Steel; Specification MIL-S-18729, condition N.

The humidified packings and the metallic plates shall be immersed in Specification MIL-H-6083, type I fluid and drained to the drip point. The packings and plates shall then be so laid together in a stack that at least two packings contact each specified metal. The stack shall be held together with a pressure of 20 to 30 pounds and placed in a desiccator which is maintained at 90 to 92 percent relative humidity at room temperature. A separate set of metallic plates shall also be preapred (buffing, cleaning, and dipping in rust-preventive fluid and drained) and placed in this desiccator in such a manner that the control plates do not touch each other or any of the packings. (Di-potassium acid phosphate, K_2HPO_4, when placed in distilled water in sufficient quantity to produce a concentrated solution, will maintain approximately 92 percent humidity in a sealed desiccator at 68 F temperature.) Time of immersion for this portion of test shall be 96 hours minimum for all Acceptance tests and 14 days minimum for all Qualification test analyses. No more than 15 minutes should elapse between the time the test samples are removed from the prehumidifying chamber and placed in the stacked condition in the second humidity chamber.

At the termination of this test, there shall be no adhesion of the packing material to the metals. There shall be only slight evidence of pitting, erosion, corrosion, or bad discoloration on the plates, as shown by the following procedure, unless metallic plates are entirely unaffected by the test:

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a. The surfaces of the plates that were in contact with the packings shall be inspected for discoloration, deposits, pitting, etc. If any exists, the surfaces of the plates shall be washed in precipitation naphtha. Deposits determined as rubber compounds or elements therefrom, which can be removed by this process and which do not occur on the separate control plates, shall be construed as adhesion.

b. If any other marks remain on the surfaces of the plates after step (a) above, the surfaces shall be lightly polished with a nonabrasive cloth buff. Any pits or eroded marks remaining after this process shall be construed to be corrosion. Discoloration or staining (marks which do not physically affect the surface of the plates and which easily wash or buff off) shall not be considered detrimental. If any doubt should arise about the presence of pitting, erosion, or corrosion on the metal plates, a microscope of approximately 10 to 15 power magnification shall be used to determine the actual condition.

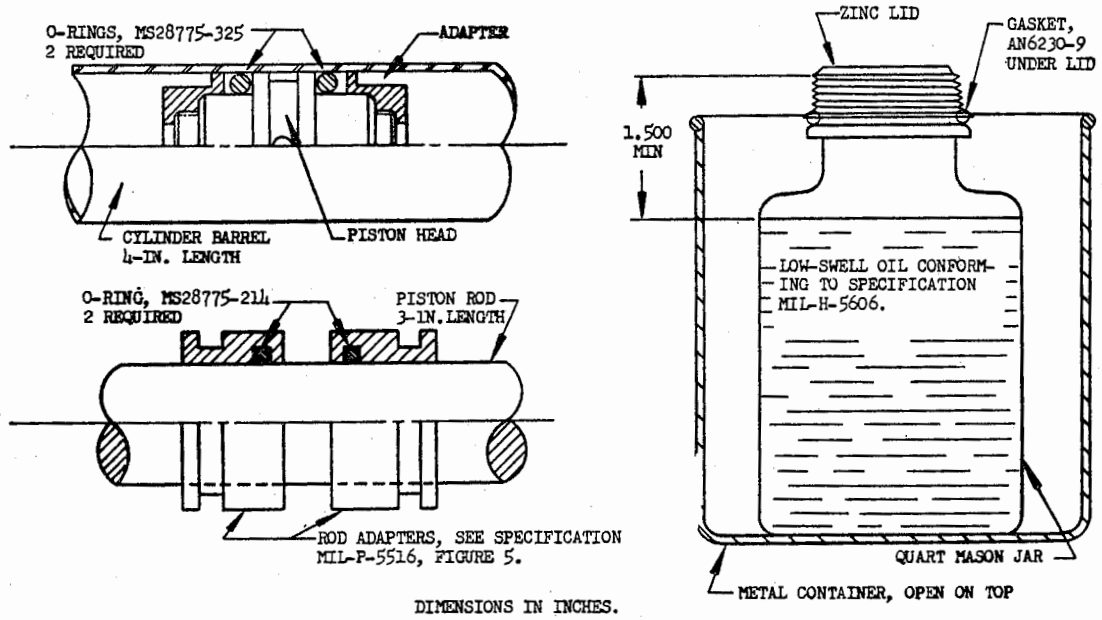
4.6.6 Performance tests "A". The following Performance tests shall be conducted in the sequence listed. All packings shall be installed in the applicable glands of test cylinder parts. These cylinder parts constitute a "dummy cylinder" (see figure 3). The dummy cylinder parts consist of a piston head having 2 glands for MS28775 -325 size packings, a cylinder barrel of approximately 4 inches in length, a 1 inch piston rod approximately 3 inches long, and 2 piston-rod packing glands for MS28775 -214 size packings. Both the piston-head and piston-rod glands shall conform to Specification MIL-P-5514. The dummy cylinder shall be placed in a quart-size mason jar and covered with oil conforming to Specification MIL-H-5606, to a minimum distance of 1½ inches from the top. A screw-type zinc lid having a glass insert shall be placed snugly on the mason jar. The jar containing the dummy cylinder shall be placed in open-top metal container. This assembly shall be put into an aging oven with the fluid temperature maintained at 275° ± 5°F for 72 hours prior to the performance tests.

All packings and the metals in contact with the packings shall be inspected at termination of this period for sticky or gummy residue, permanent set, or swelling of the packings and discoloration or corrosion of the metals. These data shall be recorded for use in conjunction with later results of the test for evaluation of performance.

4.6.6.1 Low-temperature tests "A".

4.6.6.1.1 The packings shall be removed from the dummy cylinder after aging and placed in the glands of an applicable test cylinder (see figure 4). This test cylinder shall be charged to 1,500 psi at room temperature and the accumulator pressure isolated from the test cylinder by closing the valves between the cylinder and accumulator. The cylinder pressure shall be allowed to fall to 5 to 10 psi. Temperature shall be lowered to the range of -65° to -70°F within 1 hour and held for at least 24 hours. The following cycling and static operations shall then be performed in the order specified, using Specification MIL-H-5606 fluid no warmer than -65°F:

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DIMENSIONS IN INCHES.

ASSEMBLY PROCEDURES:
PLACE DUMMY CYLINDER BARREL INTO A QUART-SIZE JAR 3/4 FILLED WITH LOW-SWELL FLUID CONFORMING TO SPECIFICATION MIL-H-5606.
INSTALL ONE MS28775-325 O-RING SEAL INTO THE GLAND GROOVE OF EACH PISTON GLAND. ADJUST THE GROOVE LENGTH TO 9/32 INCH.
INSERT THE PISTON-HEAD ASSEMBLY, WITH O-RING SEALS INSTALLED, INTO THE CYLINDER BARREL IN THE MASON JAR, WITH THE O-RINGS UNDER SQUEEZE.
INSTALL ONE MS28775-214 O-RING INTO THE GLAND OF EACH ROD ADAPTER CONFORMING TO SPECIFICATION MIL-P-5516, FIG. 5, "PERFORMANCE (AGING, HEAT AND COLD) TEST CYLINDER."
PLACE THE 2 ROD ADAPTERS, WITH O-RINGS INSTALLED, ONTO THE DUMMY PISTON ROD. PLACE THE ASSEMBLED UNIT IN THE MASON JAR. THE FLUID LEVEL SHOULD BE A MINIMUM OF 1-1/2 INCHES FROM THE TOP. PLACE AN AN6230-9 O-RING GASKET ON THE FLANGE BENEATH THE THREADED TOP OF THE MASON JAR.
SCREW A ZINC LID ONTO THE MASON JAR AGAINST THE GASKET (TIGHTLY).
PLACE THE SEALED MASON JAR INTO A METAL CONTAINER WITH THE TOP OPEN, OF AT LEAST THE SAME HEIGHT AS THE MASON JAR.
INSERT THE COMPLETE ASSEMBLY INTO A CIRCULATING-AIR OVEN SET AT 275° ±5°F FOR A 72-HOUR PERIOD.

FIGURE 3.—*Performance aging, dummy cylinder.*

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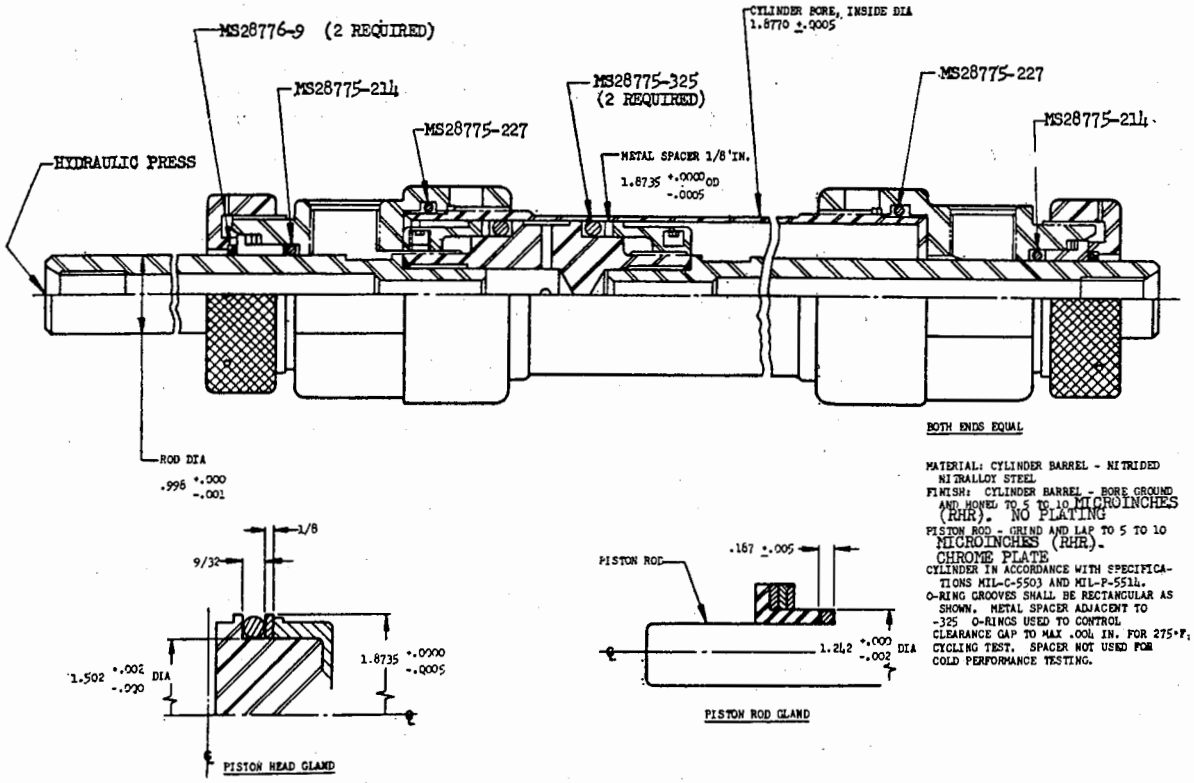


FIGURE 4.—*Cold performance and 275° F cycling test cylinder.*

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a. Low-pressure cycling: At the end of this time, the piston shall be cycled at no-load pressure for at least 10 cycles by opening the proper valves between the cylinder and reservoir and bleeding sufficient pressure from the accumulator to move the piston at the minimum pressure not to exceed 50 psi. The minimum stroke shall be 4 inches. The pressure shall be recorded. (The pressure in the cylinder shall not be permitted to exceed 50 psi at any time during these 10 cycles as by bottoming the piston, etc).

b. High-pressure cycling: The piston shall be cycled under high-pressure operation for at least 10 cycles by opening the proper valves between the cylinder and accumulator and bleeding pressure from the opposite end of the cylinder to the reservoir. Minimum cycling pressure permitted during this test is 1,500 psi.

c. High-pressure static: With the valves closed between the cylinder and the reservoir, the valves between accumulator and cylinder shall be opened to allow a static pressure of 1,500 psi minimum to both sides of piston without causing motion. The apparatus shall be allowed to set for 1 hour. Pressure shall then be reduced to 0 psi ans the apparatus allowed to warm up to room temperature over an 18-hour period.

NOTE: At the start of (a), (b), and (c), the piston rod shall be forcefully moved from side to side in each of two planes, 90 degrees apart.

Leakage during this entire series of tests shall not exceed 20 drops (approximately 1.0 cc) per gland. There shall be no evidence of wear, permanent set, extrusion of the packings, corrosion, sticking, or bad discoloration of the adapters or metals adjacent to the packings after or during the entire series of Performance tests. The leakage shall be recorded.

4.6.7 Endurance test "A".

4.6.7.1 Cycling test "A". The same packings which were used in the Performance tests "A" (4.6.6) shall be reinstalled in a test cylinder. Test conditions shall be as follows:

Oil-Specification MIL-H-5606
Temperature $275^{\circ} \pm 5^{\circ}F$.
Stroke -4 inch minimum.
Piston diametral clearance-0.004 inch.
Rod diametral clearance -0.003 inch.
Operating pressure -1,500 psi minimum.
Rate of cycling -30+5, -0 cpm minimum in thin (0.072 inch) wall cylinders.
Total cycles, minimum of 10,000 cycles at $275^{\circ} \pm 5^{\circ}F$.

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The test cylinder, as shown on figure 4, shall be mounted in an oven. Hydraulic pressure shall be applied through the hollow rod, and shall be 1,500 psi minimum in one direction of rod travel and approximately 0 psi in the opposite direction of rod travel (see figure 5).

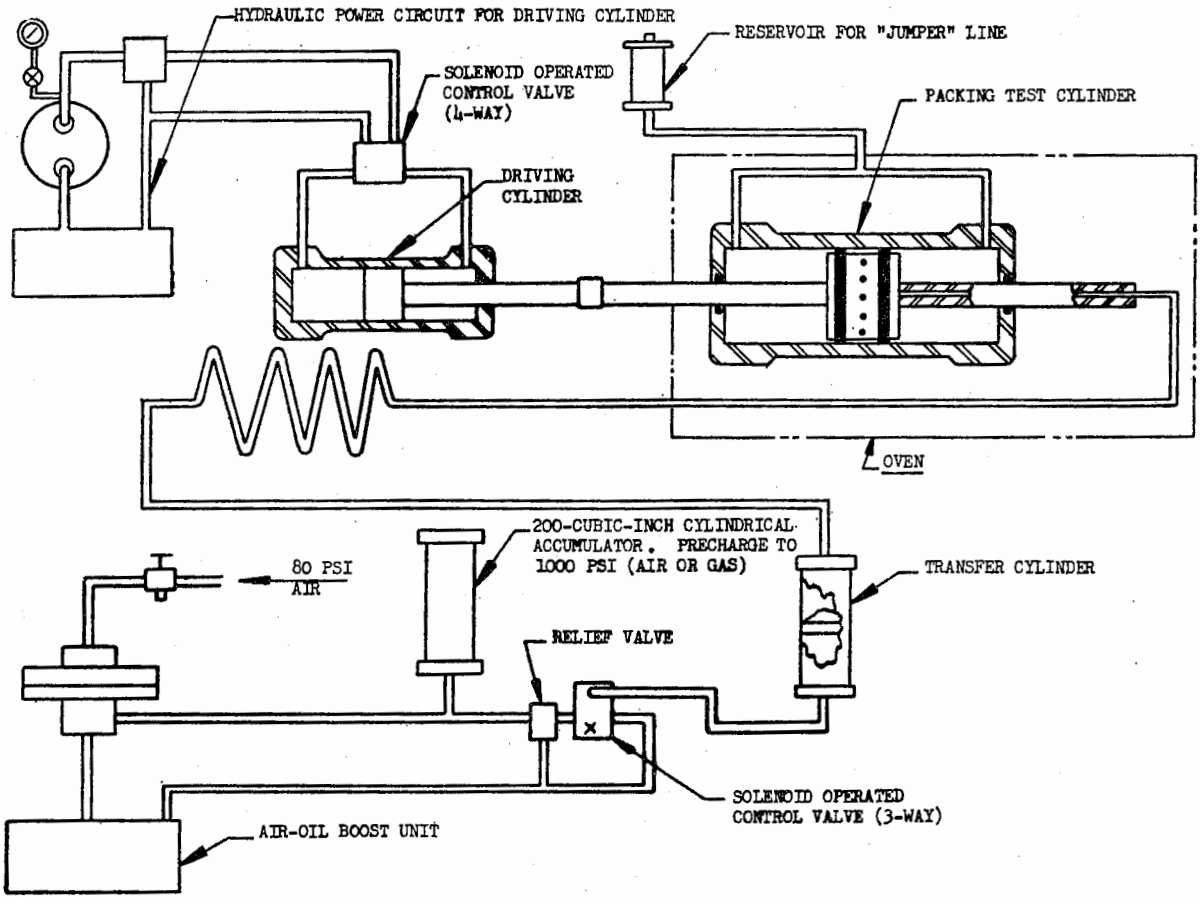


FIGURE 5.—*Endurance-cycling test circuit schematic (+275° F).*

The test cylinder shall be powered by a driving cylinder located outside of the oven. The driving circuit shall be divorced from the test pressure circuit. A "jumper" line shall join the test cylinder end-cap ports and tee into a reservoir located outside of the oven in order that fluid conforming to Specification MIL-H-5606 will always fill the test cylinder barrel. The test pressure circuit leading to the hollow rod shall contain a transfer cylinder for leakage measurement purposes.

At the completion of 10,000 cycles at 275°F, total leakage shall be determined by measuring the amount of fluid required to reposition the transfer cylinder to its starting position. (The transfer cylinder shall be bottomed at the beginning of each test.) Approximately 2,000 cycles are needed to stabilize the temperature at 275°F. A total of 12,000 continuous cycles in 1 day's cycling is required.

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4.6.8 Performance tests "B". The following performance tests shall be conducted in the sequence listed. All packings shall be installed in the applicable glands of a test cylinder similar to figure 6, and the cylinder filled with hydraulic oil conforming to Specification MIL-H-5606 at no pressure. The cylinder with packings installed shall be maintained at 518 ± 2 F for 6 days prior to performance tests. All packings and the metals in contact with the packings shall be inspected at termination of this period for sticky or gummy residue, permanent set, or swelling of the packings and discoloration or corrosion of the metals. These data shall be recorded for use in conjunction with later results of the test for evaluation of performance.

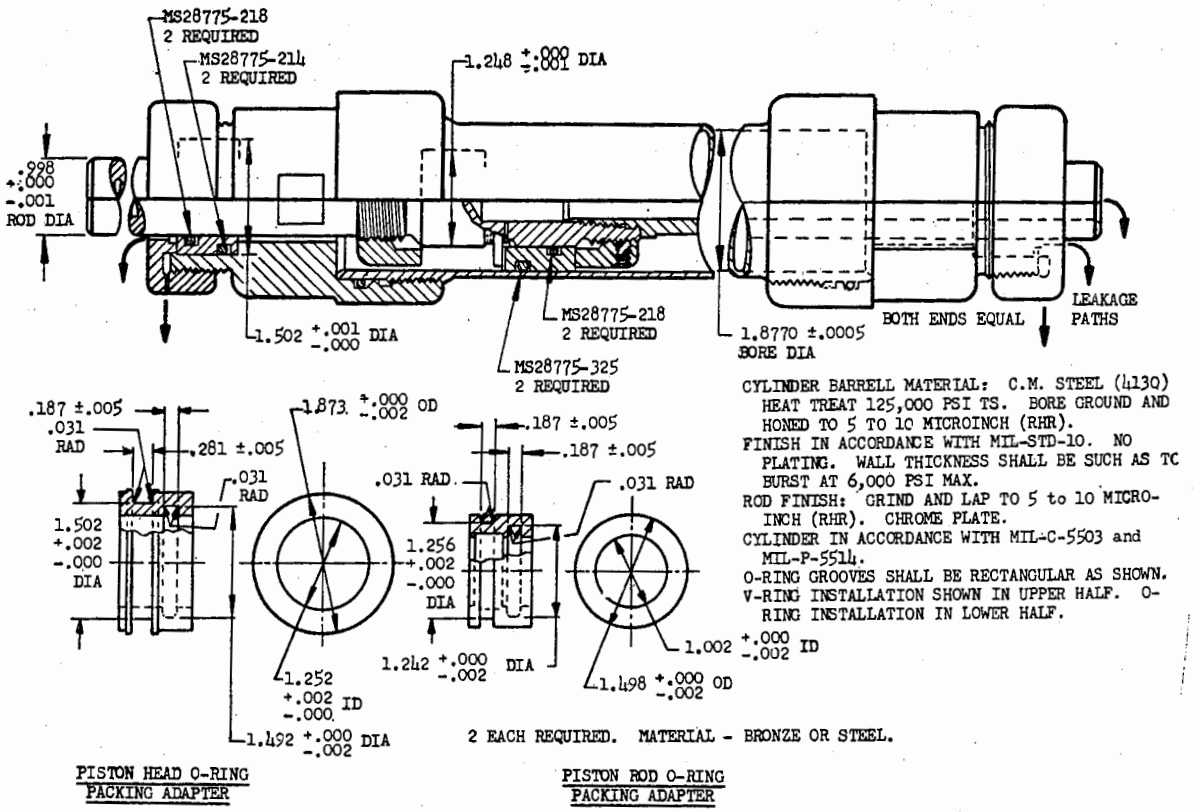


FIGURE 6.—*Performance (aging, heat and cold) test cylinder.*

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4.6.8.1 High-temperature performance.

4.6.8.1.1 High-temperature leakage. At completion of the preceding preparation, the packings shall be installed in a controlled temperature chamber as indicated in figure 7. (Only the apparatus indicated within the upper left-hand corner of figure 7 is applicable to this test.) The hydraulic fluid used for testing shall be fluid conforming to Specification MIL-H-5606. The pressure in the entire system shall be raised to 1,500 psi with the piston near one end of the cylinder, but not bottomed, and necessary valves adjusted to prevent motion. The temperature in the test box shall be raised to 160°F minimum within a period of 60 minutes, and shall remain at this temperature and pressure for 24 hours. Upon completion of the 24 hour period, the following cycling and static operations shall be accomplished in the order listed. (The specified temperature shall be maintained throughout the test.)

a. High-pressure cycling: At least 10 cycles of operation at 1,500 psi pressure shall be made by opening the proper valves between the cylinder and the accumulator and bleeding pressure from the opposite end of the cylinder to the reservoir.

b. Low-pressure cycling: At least 10 cycles of operation at low pressure shall be made by opening the proper valves between the cylinder and the reservoir and bleeding just enough pressure from the accumulator to cycle cylinder at slow motion. The pressure shall be recorded.

c. Low-pressure static: The hydraulic pressure in the test cylinder shall be lowered to 5 to 10 psi pressure and the installation allowed to set for 1 hour to determine low-pressure static leakage.

The total leakage during this entire test shall not exceed three drops (approximately 0.15 cc) from each packing gland. The leakage shall be recorded.

4.6.8.1.2 High-temperature breakout. Upon completion of the High-temperature leakage test (4.6.8.1.1), the hydraulic pressure in the entire system shall be again pumped up to 1,500 psi. With the piston held static and 1,500 psi pressure maintained, the temperature in the box shall be maintained at 160°F minimum for 3 hours, after which time the source of heat shall be turned off and the entire apparatus allowed to cool to room temperature over approximately a 20-hour period. Pressure shall not fall below 1,000 psi during this 20-hour period. A no-load breakout test shall then be performed by closing the valve between the accumulator and cylinder, exhausting the pressure in each end of the test cylinder simultaneously, and recording the lowest pressure necessary to start motion of the piston. The pressure required for breakout shall not exceed 40 psi. Leakage during this period of the test shall not exceed two drops (approximately 0.10 cc) per gland. Breakout pressure and leakage shall be recorded.

4.6.8.2 Low-temperature performance tests "B".

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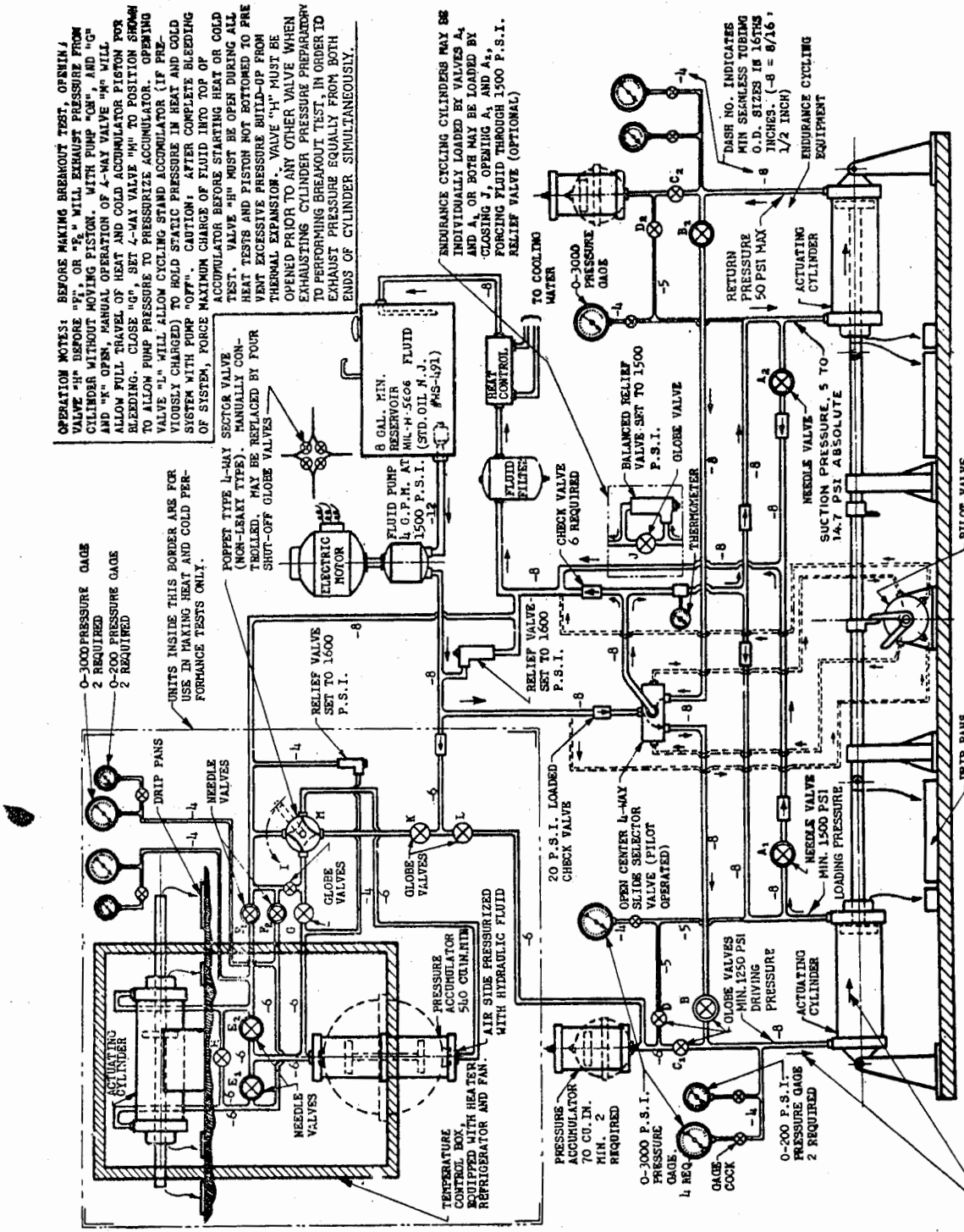


FIGURE 7.—*Packing test machine.*

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4.6.8.2.1 Leakage at low temperature. The same apparatus and packings used in previous tests shall be charged to 1,500 psi at room temperature, the accumulator pressure isolated from the test cylinder by closing the valves between the cylinder and accumulator and the cylinder pressure allowed to fall to 5 to 10 psi. Temperature shall be lowered to the range of -65° to -70°F within 1 hour and held for at least 72 hours. The following cycling and static operations shall then be performed in the order specified, using fluid no warmer than -65°F:

a. Low-pressure cycling: At the end of this time, the piston shall be cycled at no-load pressure for at least 10 cycles by opening the proper valves between the cylinder and reservoir and bleeding sufficient pressure from the accumulator to move the piston at the minimum pressure not to exceed 50 psi. The pressure shall be recorded. (The pressure in the cylinder shall not be permitted to exceed 50 psi at any time during these 10 cycles as by bottoming the piston, etc.)

b. High-pressure cycling: The piston shall be cycled under high-pressure operation for at least 10 cycles by opening the proper valves between the cylinder and accumulator and bleeding pressure from the opposite end of the cylinder to the reservoir. Minimum cycling pressure permitted during this test is 1,500 psi.

c. High-pressure static: With the valves closed between the cylinder and the reservoir, the valves between accumulator and cylinder shall be opened to allow a static pressure of 1,500 psi minimum to both sides of piston without causing motion. The apparatus shall be allowed to set for 1 hour. Pressure shall then be reduced to 0 psi and the apparatus allowed to warm up to room temperature over an 18-hour period.

NOTE: At the start of (a), (b), and (c), the piston rod shall be forcefully moved from side to side in each of two planes, 90 degrees apart.

Leakage during this entire test shall not exceed 20 drops (approximately 1.0 cc) per gland. There shall be no evidence of wear, permanent set, extrusion of the seals, corrosion, sticking, or bad discoloration of the adapters or metals adjacent to the packings after or during the entire performance tests. The leakage shall be recorded.

4.6.9 Endurance tests "B".

4.6.9.1 Cycling tests of aged ring. The same packings as were used in the Performance tests "B" (4.6.8) shall be reinstalled in a test cylinder conforming to figure 8. The cylinder shall be installed in a test apparatus conforming to figure 7. (Only the portion of figure 7 deleting temperature apparatus is applicable to this test.) This test machine shall be so constructed that during the filling stroke of the inboard end of the cylinder (at which time the piston rod is retracting), the fluid pressure shall be atmospheric or less (5 to 14.7 psi absolute pressure). Test conditions shall be as follows:

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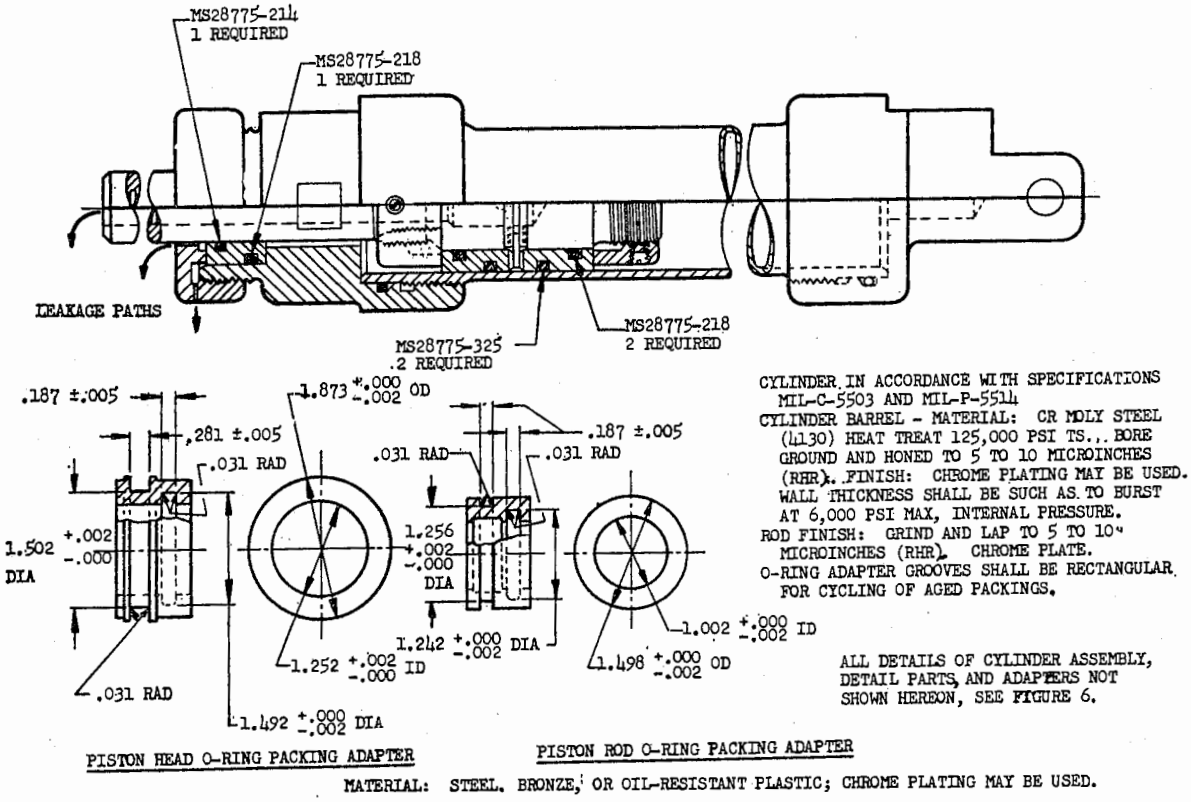


FIGURE 8.—*Endurance (aged and unaged packings) test cylinder.*

Oil-Specification MIL-H-5606
Temperature + 120° ± 10°F.
Stroke -4-inch minimum.
Operating pressure -1,100 to 1,500 psi driving, and 1,500 psi minimum loading.
Rate of cycling -30 cycles per minute minimum, total cycles, 40,000 minimum with thin-walled type cylinders. (70,000 cycles minimum if thick (non-breathing) cylinders are used.)
Standby period - Internal pressure of 1,500 psi shall be applied statically for at least a 17-hour period after every second day's cycling. Standby pressure shall be 5 to 10 psi during all other periods. In no case, however, shall more than 10,000 cycles be made between two standby periods.

All test data shall be recorded.

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4.6.9.1.1 Leakage during entire cycling test, including standby, shall not exceed an average of 0.5 cc per 1,000 cycles of operation (approximately 1 drop per 100 cycles) per gland. Leakage over the last 5,000 cycles of operation shall not exceed 10 cc. The leakage shall be recorded.

4.6.9.2 Cycling tests of new packings. A second set of new packings (as molded with no aging) shall be installed in a life-test cylinder in accordance with figure 8, as used in the Cycling tests of aged packings (4.6.9.1), and cycled under identical conditions of pressure, temperature, stroke, and speed of cycling, standby, and total number of cycles, except that the packing adapters shall be V-grooves as noted in figure 8. All test data shall be recorded.

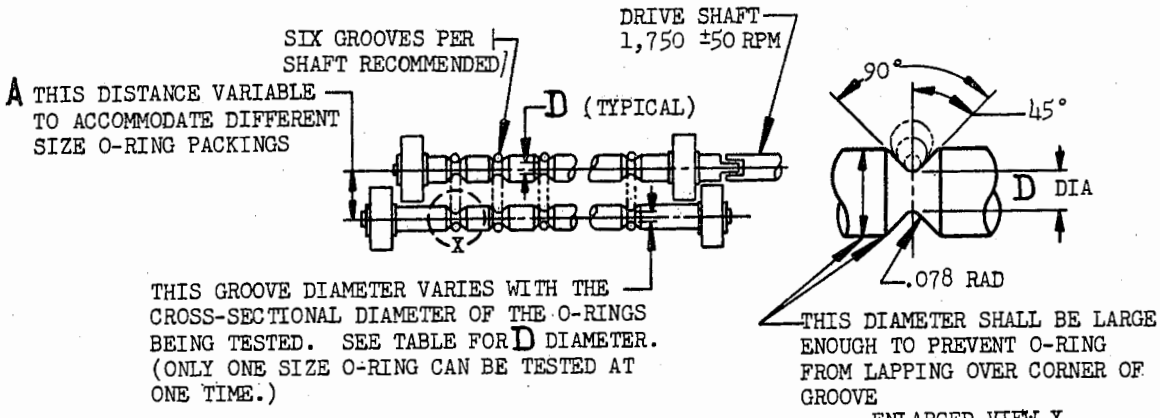
4.6.9.2.1 Leakage of new packings shall not exceed the rate allowed under endurance cycling of aged packings. The leakage shall be recorded.

4.6.9.2.2 The right is reserved by the Services to conduct cycling tests on air-aged packings in rectangular grooves and on oil-aged packings in V-grooves if additional data is considered necessary by the Services to evaluate the manufacturer's product.

4.6.10 Fatigue. Two unaged packings each of sizes MS28775-214, -218, and -325 shall be tested in a fatigue machine as described in figure 9, and the condition set up herein. The packings shall be installed in the fatigue machine on the proper size grooves as noted in figure 9, with nominal ID periphery of the packing stretched 15 percent. The driving roll shall rotate at 1,750 $\pm$ 50 rpm. All packings shall last 12 hours of rotation in the machine without signs of cracking, flaking, wear, or other deterioration, or failure. The rotation need not be continuous, but the 15-percent stretch should not be maintained during extended periods of no rotation, and the maximum test time specified should be summation of only those periods of time in which rotation takes place. The test data shall be recorded.

4.6.11 Crush. Nine MS28775-214 packings shall be chosen for this test. Six packings shall be aged in oil for a period of 168 hours at 158° $\pm$ 2°F. The oil to be used shall be designated by the Services and shall conform to Specification MIL-H-5606. Aging of the test samples shall be performed by placing the 6 test packings in a free state in a pint-size Mason jar or equivalent, containing 138.0 cc of fluid conforming to Specification MIL-H-5606. A lid should be fastened tightly on this jar to seal in all vapors and air. A gasket conforming to Part No. AN6230-9, or equivalent gasket, shall be placed under the lid as a seal. This jar shall be placed in an uncovered quart-size oil can. This assembly shall be placed in an air-circulating oven, holding the oil temperature at 158° $\pm$ 2°F for 168 hours.

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O-RING DASH NO.	STRETCH A	ROLL GROOVE DIA D
116	31/32	.250 ±.005
210		
211	1-5/64	
212	1-3/16	
213	1-19/64	
214	1-13/32	
215	1-17/32	
216	1-5/8	
217	1-3/4	
218	1-7/8	
219	1-31/32	
220	2-3/32	
221	2-13/64	
222	2-5/16	
325	2-1/32	.437 ±.005
326	2-1/4	
327	2-15/32	
328	2-45/64	
329	2-59/64	
330	3-5/32	
331	3-25/64	
332	3-39/64	
333	3-53/64	
334	4-1/16	
335	4-9/32	
336	4-1/2	
337	4-47/64	
338	4-61/64	
339	5-3/16	
340	5-13/32	

O-RING DASH NO.	STRETCH A	ROLL GROOVE DIA D
341	5-41/64	.437 ±.005
342	5-55/64	
343	6-3/32	
344		
345		
346		
347		
348		
349	7-7/16	
426	7-3/8	.625 ±.005
427		
428		
429		
430		
431		
432		
433	8-61/64	
434		
435		
436		
437		
438		
439		
440		
441		
442		
443		
444		
445	13-15/32	

EXAMPLE: MS28775-214: CIRCUMFERENCE OF NOMINAL ID (1 INCH) = 3.142 INCHES. 15 PERCENT OF 3.142 = .472 INCH. 3.142 +.472 = 3.614. CIRCUMFERENCE OF .250-INCH-DIAMETER GROOVE = .785. $\frac{3.614 - .785}{2}$ = 1.414 INCH, OR NOM SIZE 1-13/32 INCH = (DIM. A).

FIGURE 9.—*O-ring Fatigue test machine.*

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Three aged packings shall be selected at random from the 6 packings in the aged group. The aged packings selected and the 3 unaged packings shall each be twisted (inside out) ½ turn about the cross section and laid between rigid iron and steel plates with a surface finish of 32 to 63 micro-inches RHR in accordance with Standard MIL-STD-10. This may be obtained by sandblasting a commercial grind finish as indicated in figure 10. The oil-aged packings shall be laid together in one layer and the unaged packings in a second layer. The stack shall then be subjected to a constant axial compression load of 16,800 pounds or 5,600 pounds for each packing in a layer (number of layers does not affect the applied load) for the O-ring stack.

The stack shall be left in the compressed state for 20 hours and then each packing removed and inspected immediately for cracking or disintegration. The packings shall be laid away at room temperature in free condition and the permanent set of the packing cross section determined at the end of 6 hours. No cracking or disintegration shall be evident, and the cross-sectional dimensions of the packings shall have attained at least 80 percent of the "as manufactured" size at the thinnest section. This can be measured with a thin metal template.

4.6.11.1 Quality control crush tests. Quality control crush tests shall be conducted in accordance with 4.6.11, except that new unaged packings only shall be tested.

5. PREPARATION FOR DELIVERY.

See ME Letter dated 16 Feb 68 Paragraph 2.a.

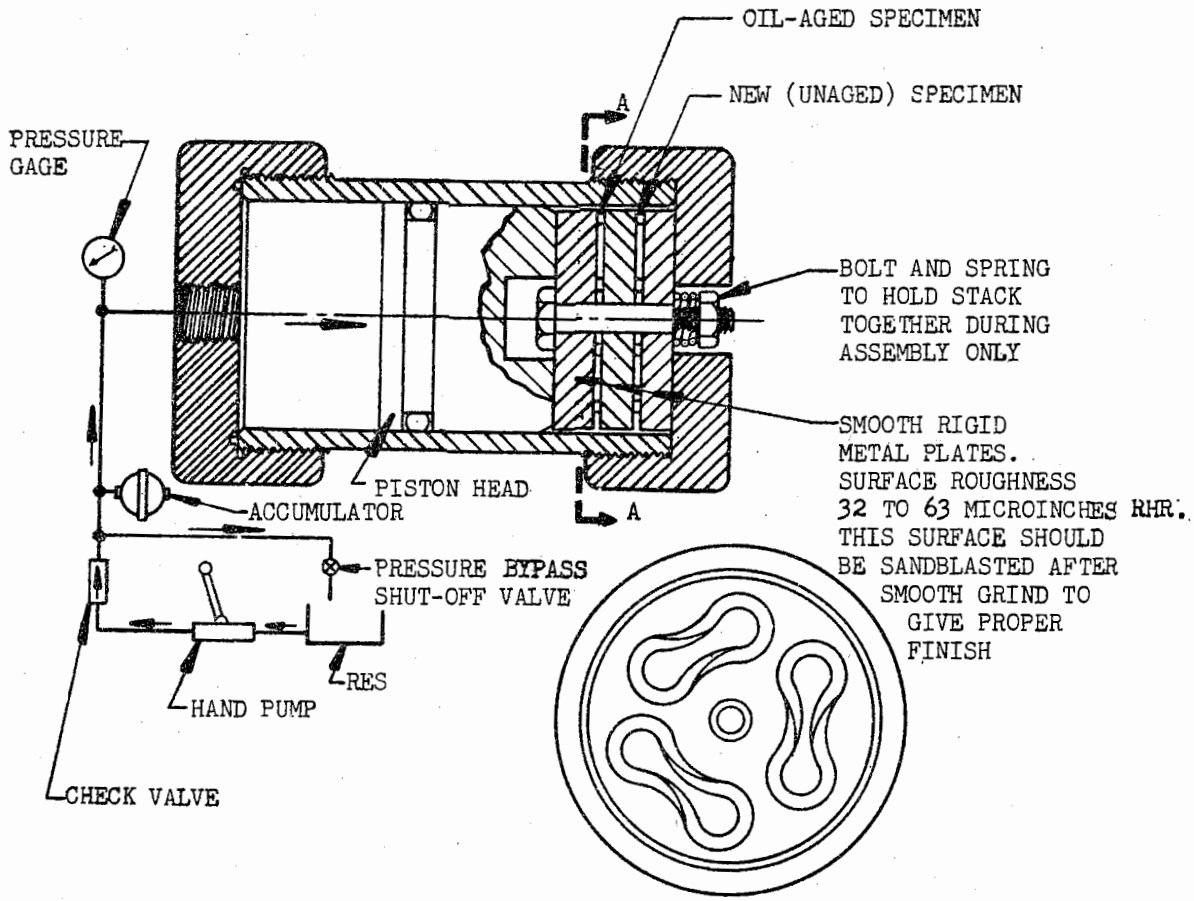
5.1 Packaging, packing and marking. Packaging, packing, and marking for the three levels of packaging and packing shall be in accordance with Specification MIL-P-4861.

6. NOTES

6.1 Intended use. This specification covers packings intended for use in packing-glands of hydraulic system components where temperatures ranging from -65° to 275°F are expected. The packings (Standard MS28775) are dimensionally interchangeable with the packings conforming to Standards AN6227 and AN6230, which are suitable for use only at temperatures ranging from -65° to 160°F. These packings should not be used in conjunction with any hydraulic fluid other than that conforming to Specification MIL-H-5606, unless otherwise specified by the procuring activity.

6.1.1 Packings are intended for use in shock-absorber struts, actuating cylinders, valves, and in all parts of hydraulic mechanisms with operating pressure of from 0 to 3,000 psi, as specified in Specification MIL-P-5514. However, diametral clearances may have to be reduced from those specified in Specification MIL-P-5514 at temperatures above 200°F.

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SUGGESTED METHOD OF MAKING CRUSH TEST. HYDRAULIC RIG SHOWN MAY BE MECHANICAL PROVIDED AXIAL CRUSHING LOAD CAN BE ACCURATELY DETERMINED AND MAINTAINED.

HYDRAULIC PRESSURE TO COMPRESS PACKINGS = $P = \frac{F \times N}{A}$

WHERE P = PRESSURE, PSI: N = NO. MS28775-214 RINGS IN EACH LAYER.
A = PISTON HEAD AREA SQUARE INCHES.
F = 5,600 POUNDS FOR MS28775-214

FIGURE 10.—*Crush test fixture.*

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. MS part number of the packing to be furnished.
c. Selection of applicable levels of packaging, and packing (see 5.1).

6.3 Volumetric swell drawing data. Within 60 days after a manufacturer's compound has been approved by the Government, a reprintable graph in Vandyke form should be submitted to the Services showing the result of an 18-day volumetric swell test in fluid conforming to Specification MIL-H-5606, at a temperature of 158 $\pm$ 2 F. Samples checked should include three packings from each cross-sectional size group on the applicable standard drawing. Packings of 0.275-inch cross-sectional diameter are not required in this test. The packings used for testing should be of the following sizes specified in Part No. MS28775-012, -110, -210, and -325. Samples of each size should be placed in a sufficiently large vessel and completely covered with fluid. Once every 24 hours throughout the first 8 days, except 1 day (for Sunday) provided this excepted day does not fall on the seventh day, and once every 48 hours thereafter until 18 days minimum are reached, each sample should be removed and the volumetric change determined and recorded. No sample should be out of the fluid for longer than ½ hour during this check, and the time of determining the check should be the same during each day of the 18-day period.

The average volumetric change of the three samples of each size tested should be shown in graph form on the same sheet plotted against time in days and volumetric change in percentage. Each line should be identified in relation to the size and type of packing. The graph should also contain the manufacturer's name, compound number, make of fluid used, and should be not greater in size than standard 8½ by 11 inches.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by the specification. The activity responsible for the Qualified Products list is MOAMA, (MONEB), Brookley AFB, Alabama, and information pertaining to qualification of products may be obtained from that activity.

6.4.1 Test report test samples, and test data. The following shall be furnished to the activity responsible for qualification:

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a. Test report: Three copies of a test report prepared in accordance with MIL-STD-831. The report shall contain a tabulation of the physical property values obtained during qualification tests. The report shall contain a complete set of photographs showing the packing as mounted for each test and showing details of any failures encountered. Diagrams of test setups shall be provided when not self-explanatory by the photographs.

b. A comprehensive description of the manufacturer's performance and endurance test machines, indicating all discrepancies between that machine and the one indicated herein, including gland dimensions, materials, and design, stroke, speed of cycling, fluid temperatures, operating pressures, etc. Exact gland dimensions should be given for each test cylinder.

Custodians:

Army - WC
Navy - WP
Air Force - 69

Preparing Activity:

Air Force - 69
Project No. 5330-0190

Review Activity:

Army - WC, MI, MO, GL
Navy - WP
Air Force - 69, 11, 71
DSA-DISC

User Activity:

Air Force
Navy

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